



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 1

1. Title : Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

3. Theory:

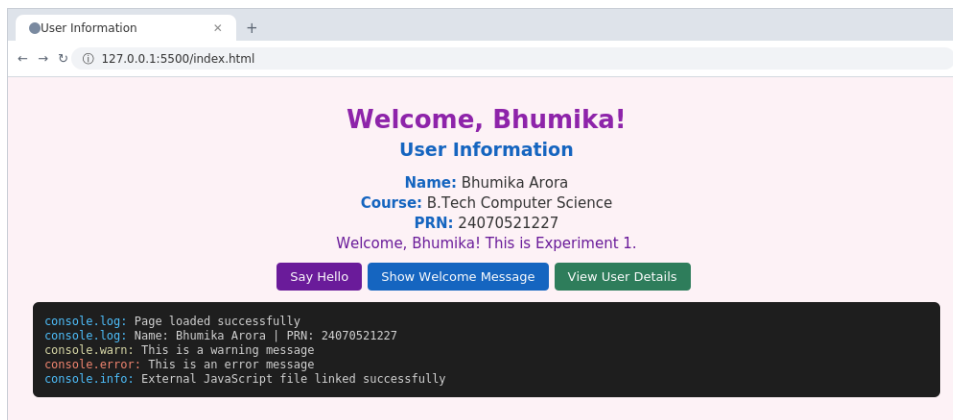
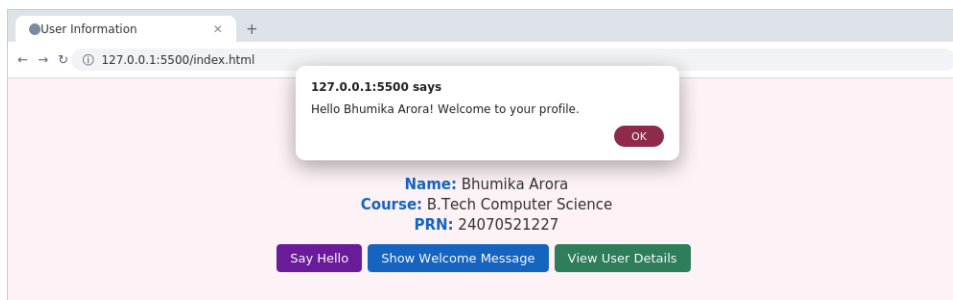
JavaScript is a lightweight, interpreted scripting language used to make web pages interactive and dynamic. It can be added to an HTML document in three different ways:

Inline JavaScript is written directly inside an HTML element using event attributes such as onclick, onmouseover or onload. In this experiment, it was used to display an alert message directly when the user clicked a button on the webpage.

Internal JavaScript is written inside the <script> tag within the same HTML file. In this experiment, JavaScript functions were defined internally and called when specific user interactions occurred on the webpage.

External JavaScript is written in a separate .js file and connected to the HTML document using the <script src="..."></script> tag. In this experiment, external JavaScript was used to keep the scripting code separate from the HTML structure, which improves readability and reusability.

4. Output:



5. Conclusion : Thus, a webpage was successfully created using HTML, CSS, and JavaScript. The three methods of integrating JavaScript, namely inline, internal, and external, were successfully implemented, and console methods were used to display the user information and welcome message through different interactive features on the webpage.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 2

1. Title : Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript (ES6).

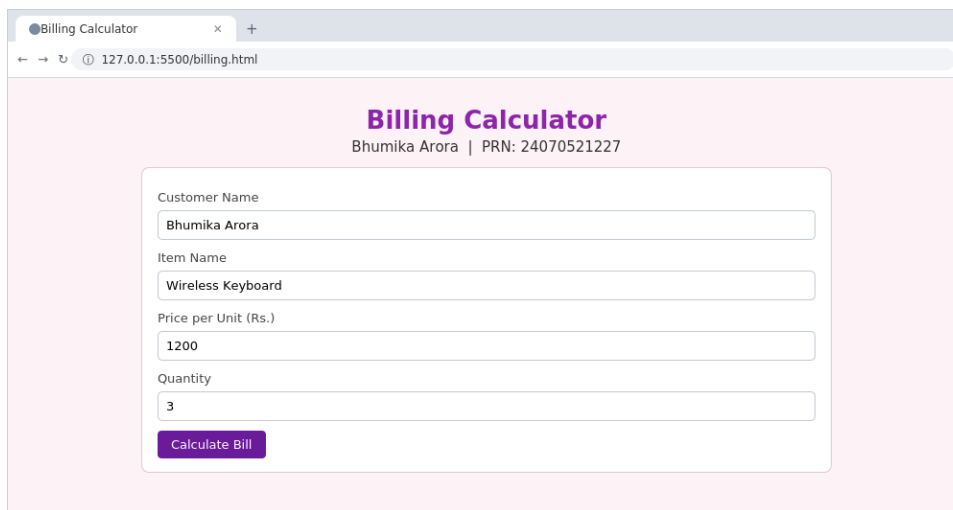
3. Theory:

Variables are containers used to store data values. ES6 introduced new ways of declaring variables along with several features that make the code shorter and easier to read.

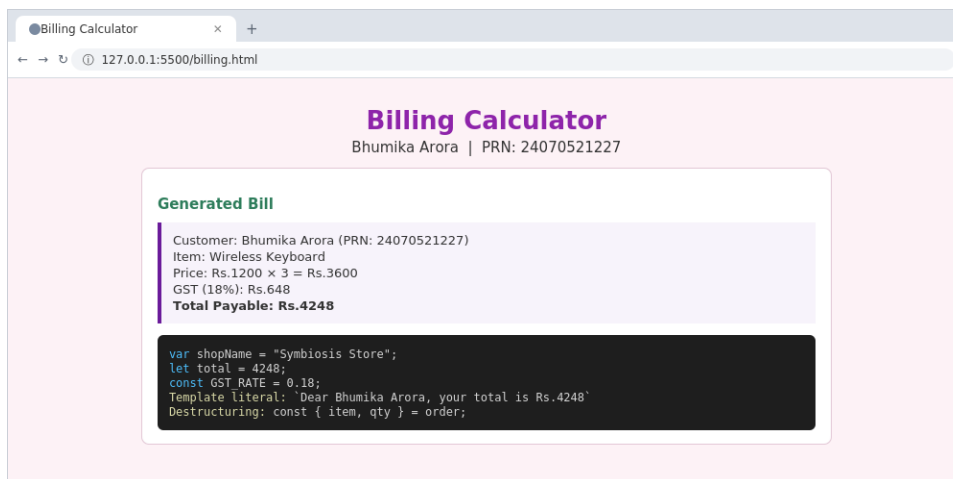
The var keyword is function-scoped and can be re-declared and updated. The let keyword is block-scoped and can be updated but not re-declared in the same scope. The const keyword is also block-scoped, but its value cannot be reassigned once initialised, which makes it suitable for fixed values such as the GST rate.

Template literals are written using backticks (` `) and allow variables and expressions to be embedded directly inside a string using the \$ {expression} syntax. They also support multi-line strings without using escape characters, and were used here to generate the formatted bill summary.

4. Output:



The screenshot shows a web browser window with the title "Billing Calculator". The address bar shows "127.0.0.1:5500/billing.html". The page has a pink background. At the top, the title "Billing Calculator" is in purple, followed by "Bhumika Arora | PRN: 24070521227". Below this is a white form with the following fields: "Customer Name" (filled with "Bhumika Arora"), "Item Name" (filled with "Wireless Keyboard"), "Price per Unit (Rs.)" (filled with "1200"), and "Quantity" (filled with "3"). A purple "Calculate Bill" button is at the bottom of the form.



The screenshot shows the same web browser window, but now the form is replaced by a "Generated Bill" section. The bill details are: "Customer: Bhumika Arora (PRN: 24070521227)", "Item: Wireless Keyboard", "Price: Rs.1200 x 3 = Rs.3600", "GST (18%): Rs.648", and "Total Payable: Rs.4248". Below the bill details is a black box containing the following code:

```
var shopName = "Symbiosis Store";  
let total = 4248;  
const GST_RATE = 0.18;  
Template literal: 'Dear Bhumika Arora, your total is Rs.4248'  
Destructuring: const { item, qty } = order;
```

5. Conclusion : Thus, the differences between var, let, and const were successfully demonstrated, and ES6 features such as template literals and destructuring were applied. A working billing calculator was built that accepts user input and displays the total amount payable along with GST.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 3

Title : Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

3. Theory: Control structures decide the order in which statements of a program are executed. They are broadly classified into conditional statements and looping statements.

Conditional Statements:

The if, else if, and else statements execute a block of code based on a condition. The switch statement compares one expression against multiple cases and is useful when there are many fixed possibilities, such as assigning a remark to each grade.

Looping Statements:

Loops repeat a block of code while a condition remains true. The for loop is used when the number of iterations is known, while loop and do-while loop are used when the number of iterations depends on a condition. In this experiment, a for loop was used to iterate over all the subjects and calculate the total marks.

Jump Statements:

The break statement exits a loop or a switch case immediately, while the continue statement skips the current iteration and moves to the next one.

4. Output:

Grading System

127.0.0.1:5500/grade.html

Student Grading System

Bhumika Arora | PRN: 24070521227

Student Name

Bhumika Arora

Marks in Web Technology (out of 100)

Please enter marks between 0 and 100.

Marks in DBMS (out of 100)

145

Invalid input: marks cannot exceed 100. Calculate Grade

Grading System

127.0.0.1:5500/grade.html

Student Grading System

Bhumika Arora | PRN: 24070521227

Subject	Marks	Grade
Web Technology	88	A
DBMS	76	B
Operating Systems	92	A+
Data Structures	64	C

Student: Bhumika Arora (24070521227)
Total: 320/400 Percentage: 80.00%
Final Grade: A — Result: PASS

```
if/else if: percentage >= 80 -> Grade A  
switch: case "A"; remark = "Excellent"  
for loop: iterated over 4 subjects
```

5. Conclusion : Thus, control structures such as if-else, switch, and loops were successfully implemented along with client-side form validation. A grading system was developed that accepts marks entered by the user, validates them, and displays the percentage and the corresponding grade.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 4

1. Title : Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

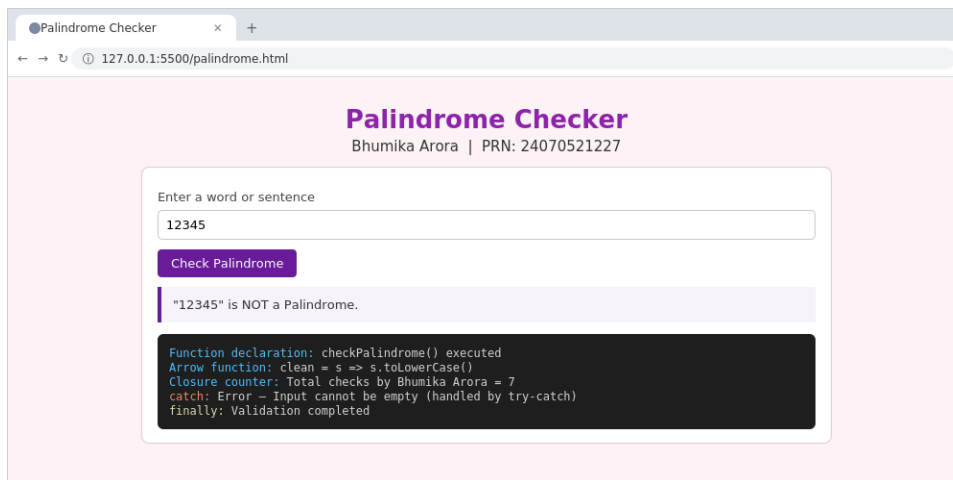
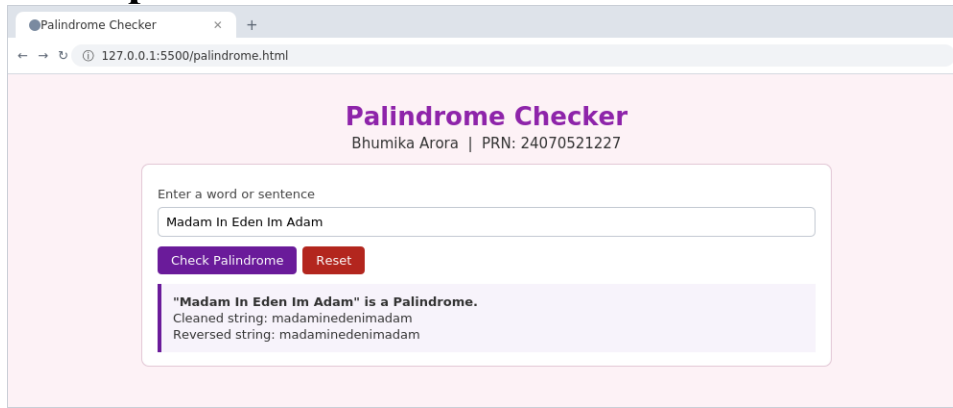
3. Theory:

A function is a reusable block of code that performs a specific task and is executed only when it is called. JavaScript supports several ways of defining functions.

Scope determines where a variable can be accessed. Global scope variables are accessible anywhere in the program, function scope variables are accessible only inside the function in which they are declared, and block scope applies to variables declared with `let` and `const` inside a pair of curly braces.

A closure is created when an inner function remembers and continues to access the variables of its outer function even after the outer function has finished executing. In this experiment, a closure was used to maintain a private counter that keeps track of how many times the palindrome check was performed.

4. Output:



5. Conclusion : Thus, different types of functions, the concept of scope, and closures were successfully demonstrated. Error handling using try-catch was applied, and a working palindrome checker was built that verifies whether the string entered by the user reads the same forwards and backwards.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 5

1. Title : Apply array methods and object handling; create a cart total calculator with discount logic.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript (ES6).

3. Theory:

An array is an ordered collection of values stored in a single variable, while an object stores data in the form of key-value pairs. Together they are used to represent real-world data such as a shopping cart.

1. Basic Array Methods:

`push()` adds an element at the end, `pop()` removes the last element, `shift()` removes the first element, `unshift()` adds an element at the beginning, and `splice()` adds or removes elements at a specific index.

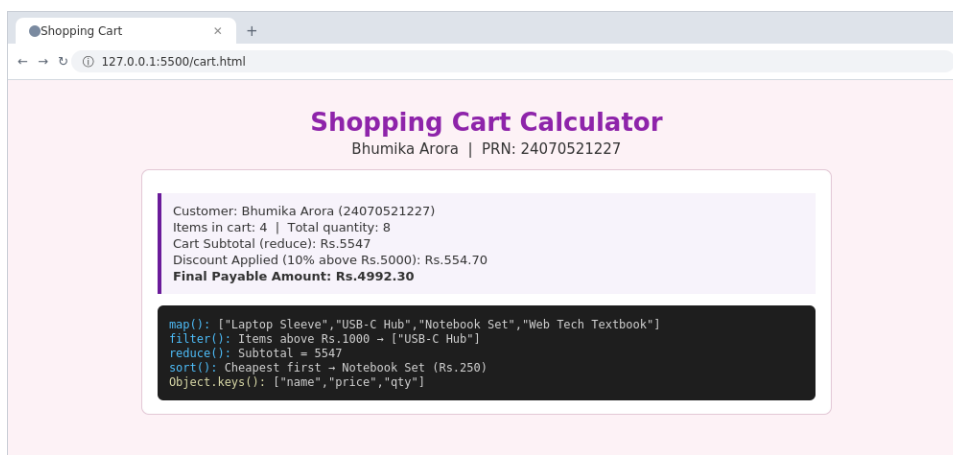
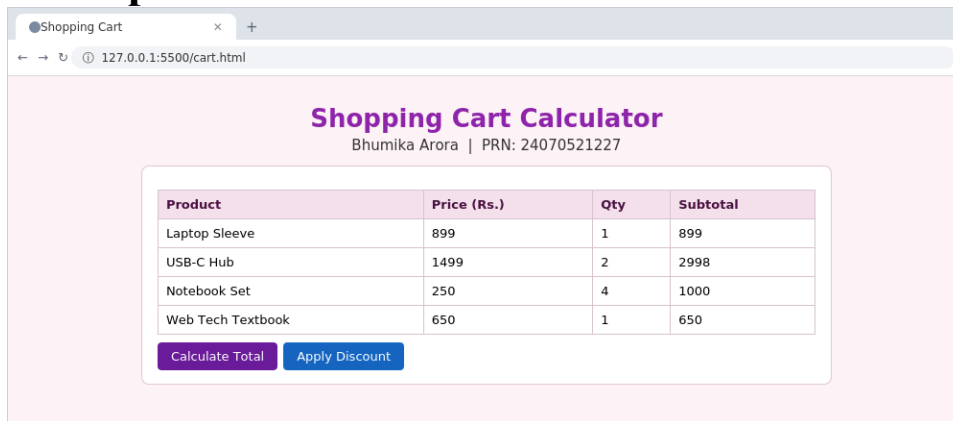
2. Higher-Order Array Methods:

`map()` creates a new array by applying a function to every element, `filter()` returns a new array containing only the elements that satisfy a condition, `reduce()` reduces the entire array to a single value, `find()` returns the first matching element, and `sort()` arranges elements in a specified order. In this experiment, `reduce()` was used to calculate the cart subtotal.

3. Object Handling:

Object properties can be accessed using dot notation or bracket notation. `Object.keys()` returns an array of the property names, `Object.values()` returns the values, and `Object.entries()` returns key-value pairs. The spread operator is used to copy or merge objects.

4. Output:



5. Conclusion : Thus, array methods and object handling techniques were successfully applied in JavaScript. A cart total calculator was created that computes the subtotal of all items using higher-order array methods and applies discount logic to display the final payable amount.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 6

1. Title : Use string functions and regex for email validation, data extraction, and text analysis.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

3. Theory:

A string is a sequence of characters, and JavaScript provides a rich set of built-in methods to manipulate strings. Regular expressions provide a powerful way to search for patterns within text.

1. String Methods:

length returns the number of characters, toUpperCase() and toLowerCase() change the case, trim() removes extra spaces, slice() and substring() extract a portion of the string, indexOf() returns the position of a character, split() converts a string into an array, and replace() substitutes matched text.

2. Regular Expressions:

A regular expression is a pattern written between two forward slashes that describes the structure of the text being searched. Meta characters such as \d for digits, \w for word characters, and quantifiers such as +, *, and {n} define how many times a pattern repeats. The flags g and i are used for global and case-insensitive matching.

3. Regex Methods:

test() returns true or false depending on whether the pattern matches, match() returns the matched results, and replace() can use a pattern to substitute text. In this experiment, test() was used to validate the email format.

4. Output:

The screenshot shows a web browser window with the title 'String & Regex' and the URL '127.0.0.1:5500/regex.html'. The page has a pink background and a purple header. The header text is 'Email Validation & Text Analysis' followed by 'Bhumika Arora | PRN: 24070521227'. The main content area contains a form with two input fields. The first field is labeled 'Enter Email Address' and contains the text 'bhumika.arora.24@sitnagpur.siu.edu.in'. Below this field is a green checkmark and the text 'Valid email address (matches regex pattern)'. The second field is labeled 'Enter another Email' and contains the text 'bhumika@gmail'. Below this field is a red 'x' and the text 'Invalid email address format.' followed by a purple 'Validate' button.

The screenshot shows the same web browser window as the previous one. The page title and URL are the same. The header text is the same. The main content area contains a form with a text area labeled 'Paragraph for Analysis' containing the text 'Bhumika Arora, PRN 24070521227, can be reached at bhumika.arora.24@sitnagpur.siu.edu.in or on 9876543210.'. Below the text area is a purple box with the title 'Text Analysis Result' and the following text: 'Total characters: 112 | Total words: 16', 'Uppercase: BHUMIKA ARORA | Lowercase: bhumika arora', 'Extracted Email: bhumika.arora.24@sitnagpur.siu.edu.in', 'Extracted Phone: 9876543210 | Extracted PRN: 24070521227', and 'slice(0,13): "Bhumika Arora" | indexOf("PRN"): 15'.

5. Conclusion : Thus, string functions and regular expressions were successfully used in JavaScript. Email addresses were validated against a defined pattern, useful data such as phone numbers and PRN were extracted from a paragraph, and basic text analysis was performed on the entered text.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 7

1. Title : Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

3. Theory: The Document Object Model (DOM) represents an HTML document as a tree of objects. JavaScript uses the DOM to read and modify the content, structure, and style of a webpage after it has loaded.

1. Selecting Elements:

`getElementById()` selects a single element by its id, `getElementsByClassName()` and `getElementsByTagName()` return collections of elements, `querySelector()` returns the first element matching a CSS selector, and `querySelectorAll()` returns all matching elements.

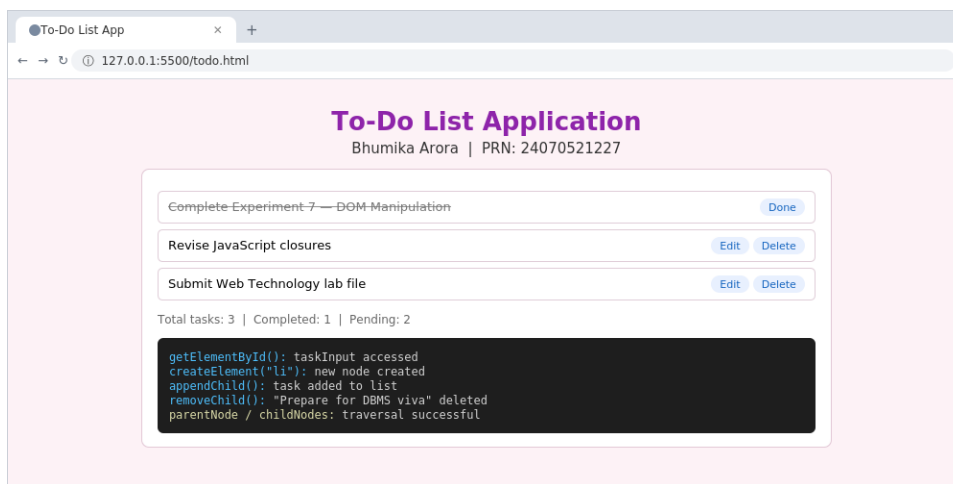
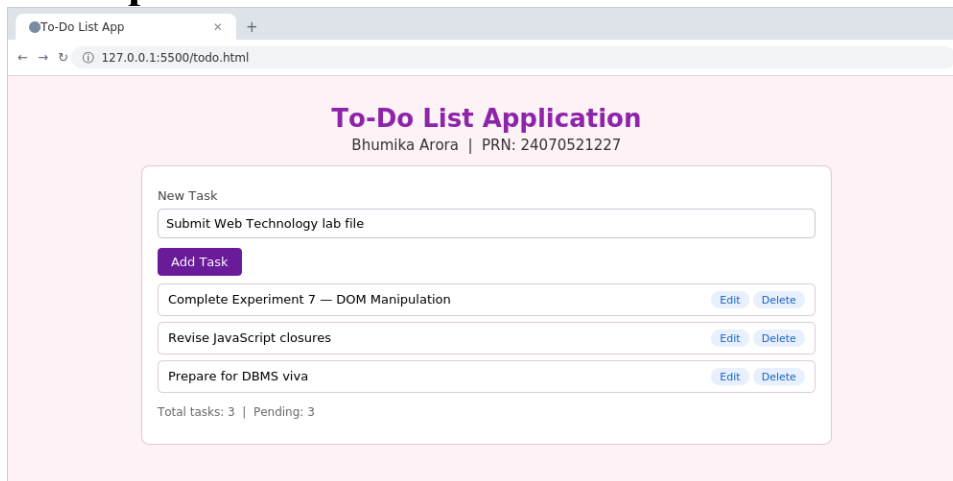
2. DOM Traversal:

`parentNode` moves to the parent of an element, `childNodes` and `children` access its child elements, and `firstChild`, `lastChild`, `nextSibling`, and `previousSibling` move between related nodes in the tree.

3. Modifying the DOM and To-Do List Implementation:

`innerHTML` sets or returns the HTML content of an element, `textContent` handles plain text, `createElement()` creates a new node, `appendChild()` inserts it into the document, `removeChild()` deletes a node, and `setAttribute()` and `classList` are used to change attributes and styles.

4. Output:



5. Conclusion : Thus, DOM traversal and DOM manipulation methods were successfully implemented in JavaScript. A fully functional to-do list application was developed in which tasks can be added, edited, and deleted dynamically without reloading the webpage.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 8

1. Title : Access and validate form fields using events; design a gym admission form with live input checks.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

3. Theory: An event is an action that occurs on a webpage, such as a click, a key press, or a change in an input field. Event handling allows JavaScript to respond to these actions and validate data as the user types.

1. Accessing Form Fields:

Form elements can be accessed using `document.forms`, `getElementById()`, or `querySelector()`. The `value` property is used to read or set the content of a field, while `checked` is used for checkboxes and radio buttons.

2. Event Handling Methods:

Events can be attached using HTML event attributes, using DOM properties such as `element.onclick`, or using `addEventListener()`, which is preferred because it allows multiple handlers to be attached to the same element.

3. Live Validation:

In this experiment, each field of the gym admission form was validated as the user typed. The name field was checked for emptiness, the PRN and mobile number were checked for the correct length and digits, and the email was verified against a regular expression, with success or error messages displayed instantly beside each field.

4. Output:

The screenshot shows a web browser window with the title "Gym Admission Form" and the URL "127.0.0.1:5500/gym.html". The form is titled "Gym Admission Form" and displays the user's name "Bhumika Arora" and PRN "24070521227". The form contains the following fields and messages:

- Full Name: (Valid, green checkmark: "Looks good")
- PRN: (Valid, green checkmark: "Valid PRN")
- Email: (Invalid, red error: "Enter a valid email address")
- Mobile Number: (Invalid, red error: "Mobile number must be 10 digits")
- Membership Plan: (Dropdown menu)
- Submit button

The screenshot shows the same web browser window after successful submission. A success message is displayed in a white box with a red "OK" button:

127.0.0.1:5500 says
Form submitted successfully!
Welcome to the gym, Bhumika Arora (24070521227).

Below the message, a purple box displays the registration details:

Registration Successful!
Name: Bhumika Arora
PRN: 24070521227
Email: bhumika.arora.24@sitnagpur.siu.edu.in
Mobile: 9876543210
Plan: Quarterly — Rs.3000
Status: Confirmed

At the bottom, a black console log displays the following JavaScript events:

```
onblur: name field validated  
oninput: live check on mobile number  
onchange: membership plan updated  
onsubmit: event.preventDefault() called
```

5. Conclusion : Thus, form fields were successfully accessed and validated using JavaScript events. A gym admission form was designed with live input checks that guide the user while filling the form and allow submission only after all the entered details are valid.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 9

1. Title : Implement localStorage and sessionStorage to save user preferences like theme settings.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, and JavaScript.

3. Theory:

Web Storage is an HTML5 feature that allows data to be stored inside the user's browser in the form of key-value pairs. It is simpler and offers far more capacity than cookies, and the stored data is never sent to the server with every request.

1. localStorage:

Data stored in localStorage has no expiry time and remains available even after the browser is closed and reopened. It is used for saving long-term preferences such as the selected theme, font size, or language.

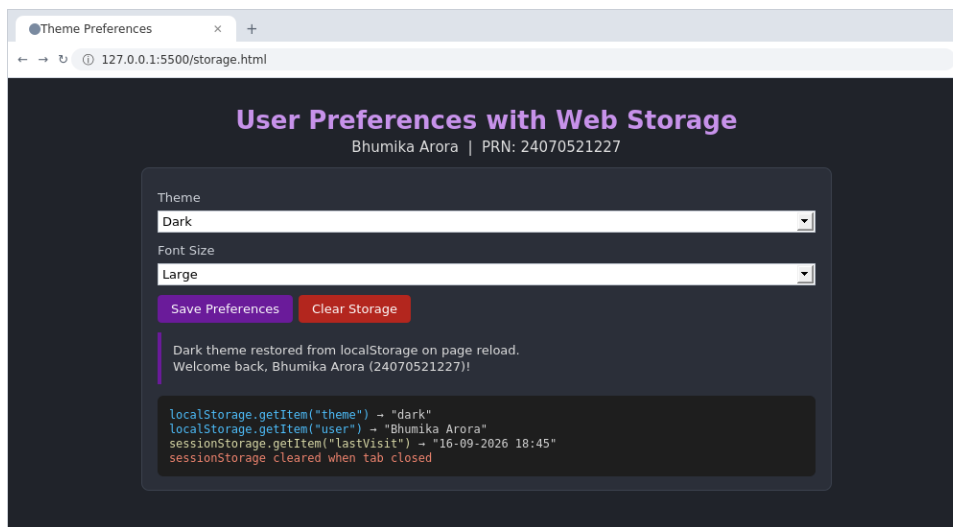
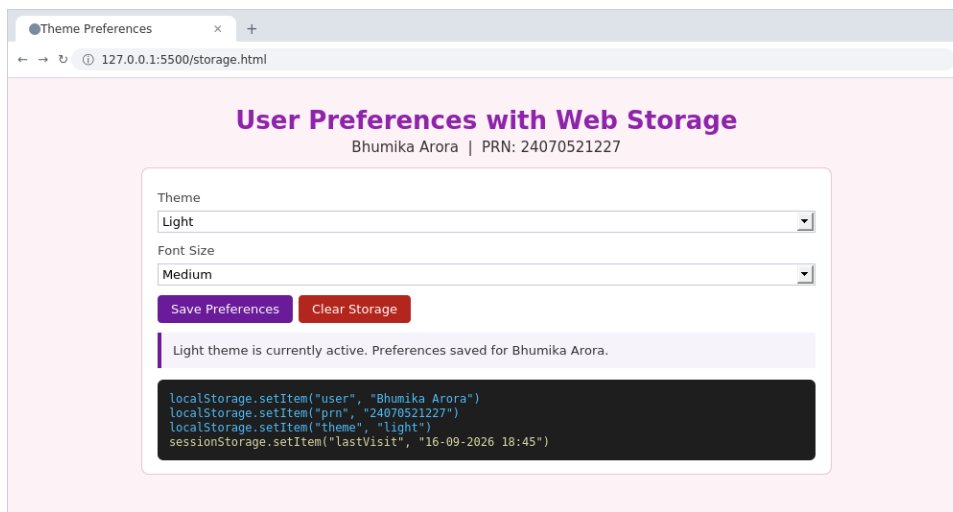
2. sessionStorage:

Data stored in sessionStorage is available only for the duration of the current browser tab or session and is automatically cleared once the tab is closed. It is suitable for temporary data such as the time of the last visit within a session.

3. Storage Methods:

setItem(key, value) stores a value, getItem(key) retrieves it, removeItem(key) deletes a specific entry, clear() removes all stored data, and the length property returns the number of stored items. Since only strings can be stored, JSON.stringify() and JSON.parse() are used for objects.

4. Output:



5. Conclusion : Thus, localStorage and sessionStorage were successfully implemented in JavaScript. User preferences such as the theme and font size were stored in the browser and automatically restored on the next visit, while the difference in the lifetime of the two storage types was clearly observed.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO: 10

1. Title : Load and display JSON data using fetch() and jQuery \$.getJSON(); render results dynamically in a table.

2. Software/Tools Required: Visual Studio Code, Google Chrome, HTML5, CSS3, JavaScript, jQuery, and Live Server.

3. Theory: JSON (JavaScript Object Notation) is a lightweight, text-based format used to exchange data between a server and a web page. AJAX allows this data to be requested and displayed without reloading the entire page.

JSON data is written as key-value pairs inside curly braces, and multiple records are stored inside square brackets as an array of objects. The keys must always be enclosed in double quotes.

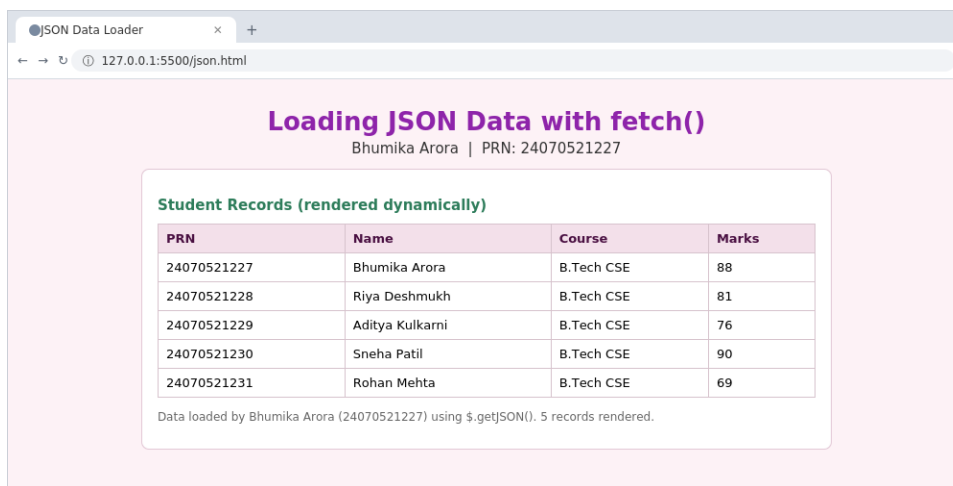
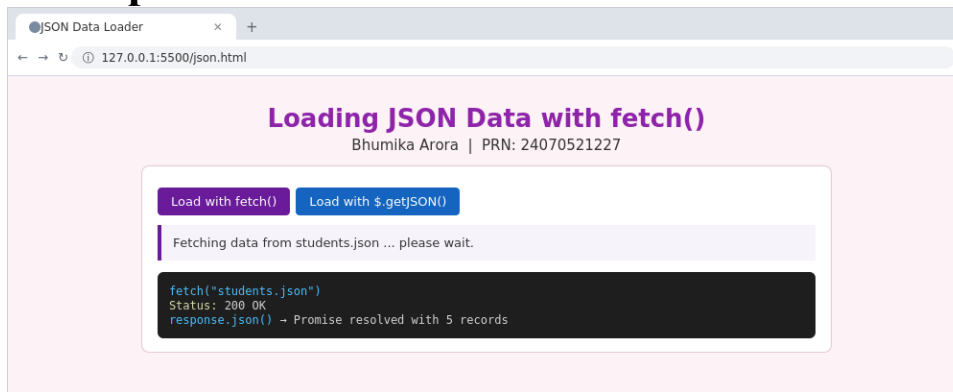
2. The fetch() API:

fetch() is a modern, promise-based method used to make network requests. It returns a promise that resolves to a response object, on which the .json() method is called to convert the body into a JavaScript object. The .then() method handles the successful result, while .catch() handles any network or parsing error.

3. jQuery \$.getJSON() and Dynamic Rendering:

jQuery provides a simpler shorthand method, \$.getJSON(url, callback), which sends an HTTP GET request and automatically parses the JSON response. The data is then available directly inside the callback function. The received array of records was looped through, and for every record a table row was generated and inserted into the table body using DOM manipulation, so that the table content was created entirely at runtime.

4. Output:



5. Conclusion : Thus, JSON data was successfully loaded using both the `fetch()` API and the jQuery `$.getJSON()` method. The retrieved records were rendered dynamically into an HTML table, demonstrating how data can be displayed on a webpage without reloading it.